

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 166

CREW VITALS WATCHES

The invention is the plumbing
detected is dispatched



the wrist reports

HR · SPO2 · FALLS · AI LINK
NEAR-IMMEDIATE, HONESTLY

Crew system — v1.0 in force

GP-IM-166

Revision v1.0 — 23 August 2026

167 of 290

VETTED BATCH 20 — PASS · WORDING FIX SEATED

PRIMARY AUTHOR & INVENTOR: ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPHD

CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 166

PHASE 166: CREW VITALS WATCHES — LIFE-SYSTEM MONITORING ON SHIP TELEMETRY — STATUS: CURRENT — ARCHITECTURE PASS; THE 'ZERO MINUTES' WORDING FIXED TO NEAR-IMMEDIATE NOTIFICATION (AUTHOR'S ORDER 'FIX THE REST', 15 AUGUST 2026). The consumer watch, promoted to ship's instrument. Phase 166 puts heart rate, SpO2, rhythm and fall detection on every wrist — consumer-pattern hardware, no invention needed — and wires it crew-wide into ship telemetry with a direct watch-to-AI link, the link being the critical piece on a planetary transit: the difference between triage as a search problem and triage as a routing problem. The phase's own physics note is seated: the sensor suite already exists — the phase is plumbing, not invention, and plumbing is what saves lives at 03:00 ship-time. The vet graded the architecture sound with one wording fix — 'zero minutes' read as near-immediate notification — adopted on the author's order 'fix the rest' and carried verbatim in §1.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-166, v1.0 — 23 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the watch law as seated (contents line, the bodies — the device, what it is for, the physics note, the provenance — plus the wording fix, verbatim) — §2 why plumbing beats invention (graded) — §3 the system on telemetry — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 166: **Crew Vitals Watches — Life-System Monitoring on Ship Telemetry**. Consumer-pattern watches (heart rate, SpO2, rhythm, falls) reporting crew-wide vitals to ship telemetry with a direct watch-to-AI link critical on planetary excursions; converts triage from a search problem into a routing problem.'

THE CORE OBJECTIVE: none seated as a standalone line — the contents line above carries the register wording.

Live instruction, 29 July 2026. Written in real time with Archer, immediately after Phase 165.

Deliberately simple: the device already exists in the consumer world; the design act is the upgrade — wiring it into ship telemetry so the fleet, not just the wearer, knows the state of every life aboard.

THE DEVICE (VERBATIM)

- **The Watch:** every crew member wears the new-style watch — the consumer pattern that already monitors life systems: heart rate, blood oxygen, rhythm, movement, falls.
- **The Upgrade:** ship telemetry. Every watch reports to the ship: who is alive, where they are, and what their body is doing — continuously, crew-wide.
- **The Direct AI Link — critical (second instruction, 29 July 2026):** every watch links directly to AI, not only through ship telemetry. On planetary excursions this becomes the primary channel: the ship may be over the horizon or in orbit, hours from a response — the AI unit standing beside you is the one that catches you.

WHAT IT IS FOR (VERBATIM)

- **Triage Intelligence:** when Phase 163's 50/40 doctrine fires, the 40% first-response AI do not search blind — the watches have already told the ship who is down, where, and how badly. Response order is computed from live vitals, not guessed from silence.
- **The Lone-Casualty Loop:** Phase 163's wall stations assume a casualty who can still reach them. The watch covers the casualty who cannot — unconscious in a sealed section, vitals collapsing, location known to the metre. The rescue comes to them.
- **Everyday Health:** a century-ship has no emergency room down the road; continuous vitals catch the slow emergencies — arrhythmia, hypoxia from a failing scrubber section, decompression sickness, radiation overexposure logging — long before they become Phase 163 events.

PHYSICS NOTE — PHASE 166 (KIMI AUDIT: AFFIRMED — THE SENSOR SUITE ALREADY EXISTS) (VERBATIM)

- **Every sensor named here is shipping consumer hardware.** Optical heart-rate, blood-oxygen saturation, single-lead ECG, accelerometer fall detection, and crash detection are all in production wrist devices today — fall detection already auto-calls emergency services on Earth. The sensor-side engineering risk of this phase is near zero — the surviving ship-wide network in a damaged vessel is the real engineering problem; the innovation is architectural, not sensory. *[WORDING FIXED 15 AUGUST 2026 — vet batch 20, the author's order 'fix the rest'; original: 'The engineering risk of this phase is near zero — too broad*

per the vet: the sensors are mature; the ship-wide redundant network is the new engineering problem']

- **The telemetry link is the design, and it must be survivable.** A vitals network that dies when the ship network dies is useless in exactly the Phase 163 scenarios it serves — so the audit's one design consideration: the watches should mesh peer-to-peer as well as ship-linked, passing vitals watch-to-watch through a damaged section until a surviving node reaches the ship's core. Redundancy doctrine, same as the power and water grids elsewhere in this file.
- **The medical payoff is compression of the golden hour.** Trauma survival curves are ruled by minutes-to-intervention; a telemetry network converts 'discovered casualty' into 'dispatched responder' near-immediately — and lets stable casualties wait, correctly, while deteriorating ones are reached first. This is the difference between triage as a search problem and triage as a routing problem. *[WORDING FIXED 15 AUGUST 2026 — vet batch 20, the author's order 'fix the rest'; original: "dispatched responder" in zero minutes — rhetorical, not literal: a responder still has to physically travel; the vet's 'near-immediate casualty notification, not zero-minute rescue' adopted']*
- **The direct AI link is correct excursion doctrine — proximity beats bandwidth.** On a planetary surface the ship's telemetry path runs through orbital mechanics: horizon, rotation, and relay gaps the mission cannot schedule around. A direct watch-to-AI link keeps the medical channel at the speed of the casualty's own heartbeat — the excursion AI becomes field medic, dispatcher, and stretcher-bearer with zero round-trip to orbit. The audit extends its mesh requirement accordingly: on excursion, watches and AI units form a local mesh independent of the ship; aboard, ship telemetry remains primary and the mesh is the backup. Two domains, one rule — the link nearest the casualty is the one that must work.

ORIGIN OF THOUGHT — PHASE 166 (VERBATIM)

Archer, 29 July 2026: "All staff will have the new style watches that monitor life systems, simply upgraded to ship telemetry."

Why it matters, in plain English: the shortest instruction in the file may be the one that saves the most lives, because it closes the last blind spot in the casualty doctrine. Phase 163 gave the ship responders; the wall stations gave the casualty equipment; Phase 165 gave everyone reach. But all of it assumed someone knows the casualty exists. The watch removes that assumption: every heartbeat aboard is already on the ship's instruments before anything goes wrong, so the moment something does, the question is never 'is anyone in there?' — only 'who do we reach first?' One sentence, and the ship stops discovering casualties and starts expecting them.

Archer, 29 July 2026 (second instruction, same day): "i think direct link to AI is critical, especially on planetary excursions."

Why the amendment matters: the first version of this phase wired every heartbeat to the ship. This one

wires it to the nearest mind that can act on it. On board, those are usually the same thing. On a new world they never are — the ship is a dot in the sky, and the only responder within the golden hour is the AI unit that walked down the ramp with you. The casualty channel must belong to whoever is close enough to touch the casualty.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE SENSOR SUITE, AFFIRMED (the phase's own note): photoplethysmography heart rate, SpO2, rhythm flags, fall detection — all shipping consumer technology, validated at planetary scale. The phase invents nothing here, and that is its strength: no novel sensor to fail qualification.

THE LINK, AFFIRMED AS THE INVENTION: crew-wide vitals on ship telemetry with a direct watch-to-AI channel converts medical response from search to routing — the same doctrine as Phase 163's pods, at sensor level. A fall detected is a responder dispatched; the notification chain is the life-saver.

THE WORDING FIX, SEATED: 'zero minutes' was rhetoric; near-immediate notification is the engineering truth — a responder still physically travels. The fix is the author's own order and rides verbatim in §1.

§3 — THE SYSTEM ON TELEMETRY

- THE WATCH: consumer-pattern — heart rate, SpO2, rhythm, falls.
- THE TELEMETRY: crew-wide vitals on the ship bus.
- THE LINK: direct watch-to-AI channel — the critical piece.
- THE DOCTRINE FIT: detection feeds the Phase 163 routing — sensor to pod in one chain.
- THE FIX: near-immediate notification, never 'zero minutes' (verbatim in §1).

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE PLUMBING DOCTRINE (the phase's own note): the invention is the connection, not the sensor.
- THE ROUTING DOCTRINE, SENSOR LEVEL: detected is dispatched — no search problem at 03:00.
- THE CONSUMER-OFF-THE-SHELF DOCTRINE: validated-by-planet hardware beats bespoke where lives ride on uptime.
- THE HONEST-WORDING DOCTRINE: 'fix the rest' — rhetoric corrected on his order, the correction carried verbatim.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Procure the watches: consumer-pattern suite; sensor accuracy benched against medical reference.
- STEP 2 — Wire the bus: crew-wide telemetry integration; latency mapped.
- STEP 3 — Prove the link: watch-to-AI channel; fall-detection-to-dispatch drill against the clock.
- STEP 4 — Drill the chain: detected to responder-moving — near-immediate measured, honestly reported.
- STEP 5 — Publish the ledger: accuracy benches, latency maps, drill times — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 163 — the casualty pods (GP-IM-163): the routing the watch feeds.
- PHASE 164 — the snap-pressure kit (GP-IM-164): what the responder carries.
- PHASE 161 — the hot-failover mesh (GP-IM-161): the AI at the other end of the link.
- PHASE 165 — the tether gauntlets (GP-IM-165): the other wrist instrument — detection here, anchoring there.

§8 — DO-NOT-BUILD

- DO NOT invent a sensor — consumer-validated hardware is the spec.
- DO NOT say 'zero minutes' — the fix is seated: near-immediate notification; a responder still travels.
- DO NOT leave the link optional — the watch-to-AI channel is the phase's critical piece.
- DO NOT let telemetry lag — a late alarm is the search problem returning.

§9 — OWED

OWED: nothing flagged beyond the wording fix, which is seated. Standing duty: sensor accuracy benches and notification-latency maps publish from §6.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-166, v1.0 — CURRENT — the 'zero minutes' wording fixed to near-immediate notification (the author's order 'fix the rest', 15 August 2026), carried verbatim in §1, seated law, master v2.1 (numerical print order). Built 23 August 2026, nineteenth of the 20-chapter 'ok next 20, that will take us past gemini' shift (the author's order, 23 August 2026). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576 and 587): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 166